

# RAG Project Instructions

Python Files, Streamlit Deployment, and Final Submission Guide

**Build a simple RAG project using Python files, not notebooks. The final submission must include a ZIP file, a GitHub repository link, and a deployed Streamlit app link.**

## Goal

Build a simple retrieval-augmented generation project that follows the lab sequence from raw documents to a deployed Streamlit assistant.

```
documents
-> preprocessing
-> chunking
-> vector representation
-> vector store
-> context retrieval
-> prompting
-> Streamlit UI
```

## Required File Structure

Your project must contain these files:

```
01_documents.py
02_preprocessing.py
03_chunking.py
04_vector_representation.py
05_create_chroma_store.py
06_retrieve_context.py
07_prompting.py
streamlit_app.py
requirements.txt
```

## API Key Rules

- Do not write your real API key inside any Python file.
- Do not upload your real .env file.
- Use Streamlit TOML secrets for deployment.

## Streamlit Secrets

In Streamlit Cloud:

Open your app.

Click Manage app.

Open Secrets.

Add the TOML values below.

```
OPENROUTER_API_KEY = "your_openrouter_key_here"
OPENROUTER_MODEL = "openai/gpt-4o-mini"
```

Your app should read the key from Streamlit secrets when deployed:

```
try:
    if not rag.OPENROUTER_API_KEY:
        rag.OPENROUTER_API_KEY = st.secrets.get(OPENROUTER_API_KEY, "")
    rag.OPENROUTER_MODEL = st.secrets.get(OPENROUTER_MODEL, rag.OPENROUTER_MODEL)
except Exception:
    pass
```

## Final Submission

Submit three items:

- ZIP file of your project
- GitHub repository link
- Streamlit deployed app link

## Final Checklist

- All required Python files exist.
- requirements.txt exists.
- Your real API key is not included in the ZIP file or GitHub repository.
- Streamlit secrets are configured in valid TOML format.
- The Streamlit app runs successfully.
- The answer uses retrieved context.
- The answer cites sources.